1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L3 –Engineering for Usability**

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3 ***Recap***

- What we learned so far
  - General introduction to the field
  - Historical evolution
  - Core concern
  - Idea of usability

This lecture – software development life cycles

4 **Introduction**

- User-centric software – core design concern
  - Design usable system (to cater to the needs and expectations of the "layman" users)

5 **What We Need**

- A "systematic" approach!

6 **Software Development Life Cycle**

- To comprehensively capture and represent design and development activities

7 ***Engineering* a Software**

- Software development life cycles – build software in stages

8 ***Engineering* a Software**

- Example – let us try to create a calendar app

- What should we do?

9 **Calendar App - A Possible Design**

- Create a grid-like structure typically found on a physical calendar

- Put headings on each cell in the grid (representing the name of the month)

10 **Calendar App - A Possible Design**

- Create sub-grids in each cell to hold the dates

- Render the entire structure on the screen

11 **Calendar App - A Possible Design**

- Highlight the current date and month

12 **One Possible Design**

13 **Another Possible Design**

14 **Design Alternatives**
- There can be many possibilities
  - Challenge - how to chose the right one
- Require a systematic approach
  - SDLCs help

15 **SDLC**
- There are many
  - Waterfall model
  - Spiral model
  - Evolutionary model
  - …

16 **Waterfall Model**

17 **Waterfall Model**
- Most well-known SDLC model (although rarely used in practice)

18 **Waterfall Model (Classical)**
- SEVEN major stages
  - Feasibility study
  - Requirement gathering, analysis and specification
  - System design
  - Coding (implementation) and unit testing
  - Integration and system testing
  - Deployment
  - Maintenance

19 **Waterfall Model**
- The stages are depicted as a "waterfall" (hence the name)

20 **Waterfall Model (Iterative)**
- We can have iterations between the stages as well

21 **Waterfall Model (Iterative)**

22 **Spiral Model**

23 **Spiral Model - Characteristics**
- A meta model – encompasses other SDLCs

24 **Spiral Model**

25 **Spiral Model - Characteristics**
- Multiple cycles (iterations)
- Each spiral – one iteration

- Each iterations – divided into 4 phases (quadrants)

26 **Spiral Model - Quadrants**

- 1st quadrant – identify objectives & risks
  - During the first quadrant, objectives of the iterative phase are identified
  - Also the risks associated with these objectives

27 **Spiral Model - Quadrants**

- 2nd quadrant - risk assessment and mitigation
  - Identified risks analyzed in details
  - Steps taken to reduce the risks
    - E.g., if there is a risk of inappropriate requirement specification, a prototype system may be developed

28 **Spiral Model - Quadrants**

- 3rd quadrant - development
  - Develop product (prototype) after resolving identified risks and evaluate

29 **Spiral Model - Quadrants**

- 4th quadrant – evaluation and planning
  - Review results achieved so far with customer and plan next iteration around the spiral

30 **Spiral Model**

- Through the iterations, progressively more complete version of the software gets built

31 **Interactive System and SDLC**

- Interactive systems should be "usable"
  - Requires users to take into account in every stage of the design
  - Brings in lots of iteration

32 **Interactive System and SDLC**

- Difficult to express with traditional SDLCs (e.g., waterfall model)

- In the next lecture, we shall learn about an "iterative" SDLC for interactive systems

33 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 2
- Roger S Pressman (2015). Software Engineering: A Practitioner's Approach, 8th ed, McGraw-Hill Education, New York, Chapters 2-4